RGV-System

RGV Multi-Tech Services — Business Operations Management System

Service Bookings • Inventory Management • Equipment Borrowing

Web Portals • Flutter Android App • AI Assistance

Tech Stack: Laravel 12 • PHP 8.2+ • Tailwind CSS • Alpine.js • Flutter

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1. What the System Is All About

RGV-System is a complete business operations management platform built for **RGV Multi-Tech Services**. It handles the full lifecycle of service bookings, inventory, and equipment borrowing across three user experiences — a **public web portal**, an **employee web portal** with a companion **Flutter Android app**, and a powerful **admin control center**. All surfaces share one Laravel backend, one database, and one set of business rules, so data stays consistent whether it is entered from the public booking form, the employee dashboard, or a phone in the field.

> **In one sentence:** customers request services and track them publicly, employees manage their assigned work and borrow equipment (web + Android), and administrators run approvals, inventory, reporting, users, backups, and system settings from a single dashboard.

Who Uses It

Role	Platform	Key Functions
Public	Web (`/`)	Submit and track service bookings and service requests through the public portal; real-time status via reference number.
Employee	Web portal + Flutter Android app	Manage assigned bookings on the web; use the app for inventory lookup, borrow requests, returns, and account settings.
Admin	Web (`/admin`)	Full control — bookings, inventory, users, reports, backups, settings, audit trails, and import/export workflows.

2. Core Modules & Features

2.1 Booking Management

Full service-request lifecycle: **Public submits → Admin approves/rejects → Assign employee → Completed**, with real-time tracking by reference number and file attachments.

- Create booking requests from the public interface
- Track booking status in real-time via unique reference number
- Admin approval / rejection workflows with reasons
- Assign bookings to employees; calendar view for scheduled work
- Booking cancellation and completion; PDF export of booking details

2.2 Inventory Management

Track physical items with categories, stock levels, permanent QR codes, and automatic low-stock alerts.

- Create and manage inventory items with categories and specifications
- Monitor stock levels with minimum-threshold low-stock alerts
- Permanent QR code generation and public QR inventory lookup
- Inventory reports, analytics, and stock movement tracking

2.3 Borrow Request System

Employees request items → Admin approves → Items marked borrowed → Returned with condition report → Stock restored.

- Android app: scan a QR code or enter an item code to request items
- Pending requests reserve stock until approved or cancelled
- Returns: select borrowed items, report condition, choose return location
- Full history of all borrow transactions per employee
- Web and mobile enforce the same Laravel data and business rules

2.4 Users & Roles

- Spatie RBAC with **Admin** and **Employee** roles and granular permissions
- Multi-factor authentication (TOTP authenticator + email codes)
- Admin impersonation, login history, and force-logout
- User status management (active/inactive) and profile management

2.5 Notifications

14 notification types across email and in-app channels, with a bell icon using real-time polling and per-user preferences.

- Booking events: approved, rejected, assigned, completed, cancelled
- Borrow events: approved, rejected, borrowed, returned
- Low-stock alerts, new-request alerts to admins, user status changes

2.6 Reports & Analytics

- Bookings, inventory, borrow requests, and user activity reports
- Export to **PDF / Excel / CSV / JSON** with background queuing for large datasets
- AI-powered insights, booking forecasts, and inventory recommendations
- Custom date-range filtering and trend analysis

2.7 Import / Export

- Bulk CSV/Excel import for inventory and users with duplicate-handling strategies
- Multi-format export with queued background jobs

2.8 Settings

Seven-tab admin panel: **Branding, Email (SMTP), Security (password rules, login limits), Backup (scheduled + manual), Notifications, Maintenance mode, and API rate limits**.

2.9 Backups

- Spatie backup package: full and DB-only scheduled backups plus manual trigger
- SHA-256 integrity verification, downloadable ZIP files, retention policies

2.10 Audit & Trash

- Tamper-evident audit trail with an HMAC-SHA256 checksum chain
- Soft-delete trash system with restore / force-delete for 5 entity types

2.11 AI Integration

- Chatbot with provider fallback chain: **Ollama (local) → Gemini → rule-based fallback**
- AI report generation for insights, forecasts, and recommendations
- Context-grounded responses using live system aggregates

2.12 Employee Android App (Flutter)

- Employee login with MFA; Sanctum bearer-token API access
- QR scanning and manual item-code lookup for inventory
- Borrow requests, cancellations, and location-based returns
- Dashboard, borrow history, and account settings

3. System Architecture

The application is a role-based business platform with a shared backend and distinct user experiences. Requests flow through routes to controllers, pass through authentication and authorization middleware, execute business logic against Eloquent models, and return rendered Blade views or JSON — with notifications dispatched through the queue.

Layer	Technology
Backend	Laravel 12, PHP 8.2+
Database	SQLite locally (configurable to PostgreSQL / MySQL)

Authentication	Laravel Breeze, Spatie Permission (RBAC), MFA (TOTP + email), Sanctum bearer tokens for mobile
Admin / Public Web	Tailwind CSS 3, Alpine.js 3, Blade templates
Employee Mobile	Flutter (Android) with QR code scanning
PDF / Excel / QR	barryvdh/laravel-dompdf, maatwebsite/laravel-excel, simplesoftwareio/simple-qrcode
Backups / Logging	spatie/laravel-backup, spatie/laravel-activitylog + custom tamper-evident audit
AI	openai-php/client, Google Gemini, Ollama (local)
Testing	PHPUnit 11

Mobile API

The Flutter app talks to `/api/v1/mobile`, protected by Sanctum and an active-employee middleware — it never connects to the database directly. Login and MFA issue bearer tokens; protected routes require an active account with the `employee` role. The public REST API uses tiered rate limits (30/60/300/1000 requests per minute, configurable in Site Settings); mobile login is limited to 20/minute and protected employee mobile routes to 120/minute.

4. Key Numbers

Metric	Count
Application tables	38
Eloquent models	22
Migration files	28
Controllers	42
Notification classes	14
JSON API endpoints	34+
Admin routes	60+

Middleware	9

5. Security & Operations Highlights

- **Authentication:** Laravel Breeze with hashed passwords, MFA (TOTP + email), and session management
- **Authorization:** Spatie role- and permission-based access control across admin/employee surfaces
- **Protection:** CSRF tokens, SQL-injection prevention via Eloquent, input validation through form requests
- **Auditability:** tamper-evident audit trail (HMAC-SHA256 checksum chain) plus activity logging
- **Resilience:** scheduled and manual backups with SHA-256 integrity verification and retention policies
- **Rate limiting:** tiered public API limits and stricter limits on mobile login and protected mobile routes
- **Maintenance:** maintenance mode, queue-driven notifications, and config caching

6. Quick Start

```
# Clone and install
git clone <repo-url> && cd rgv-system

# Automated setup (installs deps, creates .env, generates key, migrates, builds assets)
composer run-script setup

# Or manual setup
composer install
cp .env.example .env
php artisan key:generate
php artisan migrate --force
npm install && npm run build

# Start dev environment (server, queue, logs, vite)
composer run dev
```

Mobile: `run php artisan serve --host=0.0.0.0 --port=8000`, then from `mobile/` run `flutter run --dart-define=API_BASE_URL=http://10.0.2.2:8000/api/v1/mobile` (Android emulator host).

7. Documentation Set

Document	Description
RGV-System.md	Full system documentation — models, controllers, routes
RGV-System-Overview.md	Architecture, modules, API summary
RGV-Database-Schema.md	Application schema, including mobile tokens and return evidence
RGV-Technical-Documentation.md	Architecture, security, web/mobile APIs, and middleware
RGV-System-API.md	Web and mobile API reference
RGV-Mobile-API.md	Mobile endpoint contract and deployment configuration
RGV-User-Manual.md	User guide for admin, employee web/mobile, and public workflows
mobile/README.md	Flutter app setup and verification

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